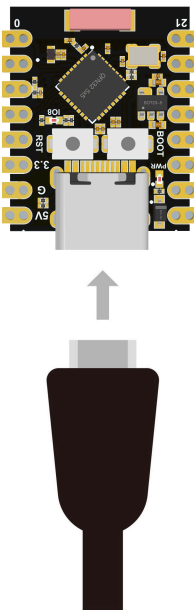


ESP32C3SuperMini 蓝牙使用

wmnologo 大约 2 分钟

硬件连接

- 步骤 1.通过USB Type-C数据线将ESP32C3SuperMini连接到计算机



扫描蓝牙

我们将使用 ESP32C3SuperMini 扫描其周围可用的蓝牙设备。

- 步骤 1.将以下代码复制并粘贴到 Arduino IDE 中

```
1  #include <BLEDevice.h>
2  #include <BLEUtils.h>
3  #include <BLEScan.h>
4  #include <BLEAdvertisedDevice.h>
5
6  int scanTime = 5; //In seconds
7  BLEScan* pBLEScan;
```

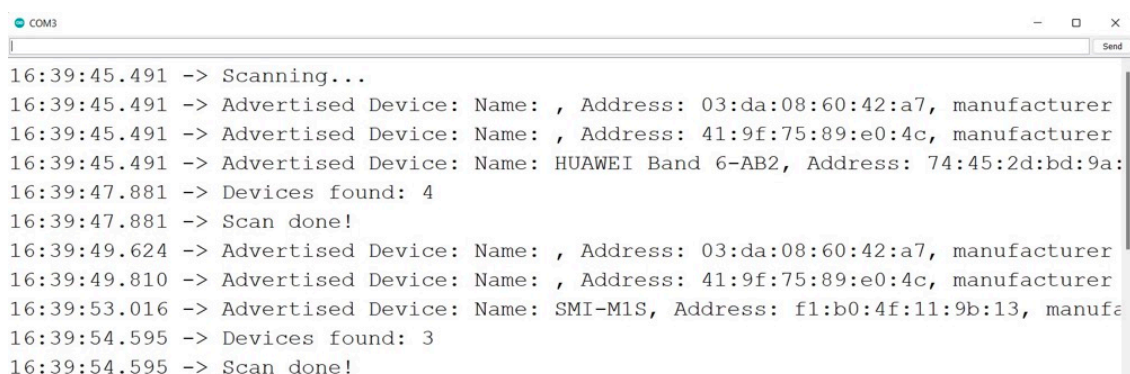
js

```

8
9 class MyAdvertisedDeviceCallbacks: public BLEAdvertisedDeviceCallbacks {
10     void onResult(BLEAdvertisedDevice advertisedDevice) {
11         Serial.printf("Advertised Device: %s \n",
12 advertisedDevice.toString().c_str());
13     }
14 };
15
16 void setup() {
17     Serial.begin(115200);
18     Serial.println("Scanning...");
19
20     BLEDevice::init("");
21     pBLEScan = BLEDevice::getScan(); //create new scan
22     pBLEScan->setAdvertisedDeviceCallbacks(new
23 MyAdvertisedDeviceCallbacks());
24     pBLEScan->setActiveScan(true); //active scan uses more power, but get
25 results faster
26     pBLEScan->setInterval(100);
27     pBLEScan->setWindow(99); // less or equal setInterval value
28 }
29
30 void loop() {
31     // put your main code here, to run repeatedly:
32     BLEScanResults foundDevices = pBLEScan->start(scanTime, false);
33     Serial.print("Devices found: ");
34     Serial.println(foundDevices.getCount());
35     Serial.println("Scan done!");
36     pBLEScan->clearResults(); // delete results fromBLEScan buffer to
37 release memory
38     delay(2000);
39 }

```

步骤 2.上传代码并打开串行监视器以开始扫描 蓝牙设备



```

16:39:45.491 -> Scanning...
16:39:45.491 -> Advertised Device: Name: , Address: 03:da:08:60:42:a7, manufacturer
16:39:45.491 -> Advertised Device: Name: , Address: 41:9f:75:89:e0:4c, manufacturer
16:39:45.491 -> Advertised Device: Name: HUAWEI Band 6-AB2, Address: 74:45:2d:bd:9a:
16:39:47.881 -> Devices found: 4
16:39:47.881 -> Scan done!
16:39:49.624 -> Advertised Device: Name: , Address: 03:da:08:60:42:a7, manufacturer
16:39:49.810 -> Advertised Device: Name: , Address: 41:9f:75:89:e0:4c, manufacturer
16:39:53.016 -> Advertised Device: Name: SMI-M1S, Address: f1:b0:4f:11:9b:13, manufa
16:39:54.595 -> Devices found: 3
16:39:54.595 -> Scan done!

```

作为蓝牙服务器

在此示例中，我们将使用ESP32C3SuperMini作为蓝牙服务器。在这里，我们将使用智能手机搜索 ESP32C3SuperMini 板并发送字符串以显示在串行监视器上

- 步骤 1.将以下代码复制并粘贴到 Arduino IDE 中

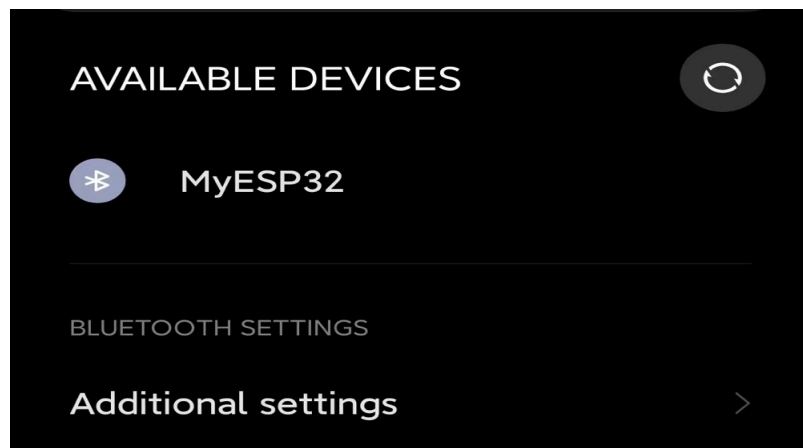
```
1  #include <BLEDevice.h>
2  #include <BLEUtils.h>
3  #include <BLEServer.h>
4
5  // See the following for generating UUIDs:
6  // https://www.uuidgenerator.net/
7
8  #define SERVICE_UUID          "4fafc201-1fb5-459e-8fcc-c5c9c331914b"
9  #define CHARACTERISTIC_UUID  "beb5483e-36e1-4688-b7f5-ea07361b26a8"
10
11
12  class MyCallbacks: public BLECharacteristicCallbacks {
13      void onWrite(BLECharacteristic *pCharacteristic) {
14          std::string value = pCharacteristic->getValue();
15
16          if (value.length() > 0) {
17              Serial.println("*****");
18              Serial.print("New value: ");
19              for (int i = 0; i < value.length(); i++)
20                  Serial.print(value[i]);
21
22              Serial.println();
23              Serial.println("*****");
24          }
25      }
26  };
27
28  void setup() {
29      Serial.begin(115200);
30
31      BLEDevice::init("MyESP32");
32      BLEServer *pServer = BLEDevice::createServer();
```

```

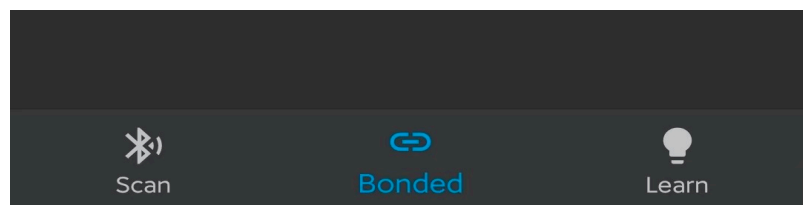
33
34     BLEService *pService = pServer->createService(SERVICE_UUID);
35
36     BLECharacteristic *pCharacteristic = pService->createCharacteristic(
37                                     CHARACTERISTIC_UUID,
38                                     BLECharacteristic::PROPERTY_READ
39     |
40
41     BLECharacteristic::PROPERTY_WRITE
42                                     );
43
44     pCharacteristic->setCallbacks(new MyCallbacks());
45
46     pCharacteristic->setValue("Hello World");
47     pService->start();
48
49     BLEAdvertising *pAdvertising = pServer->getAdvertising();
50     pAdvertising->start();
51 }
52
53 void loop() {
54     // put your main code here, to run repeatedly:
55     delay(2000);
56 }

```

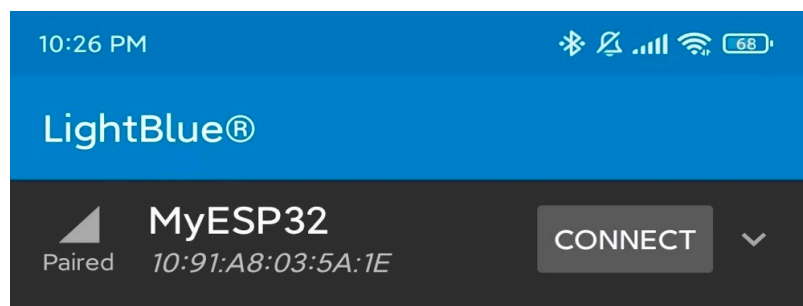
- 步骤 2.上传代码并打开串行监视器
- 步骤 3.在您的智能手机上下载并安装 LightBlue 应用程序
 - [LightBlue App \(Android\)](#)
 - [LightBlue App \(IOS\)](#)
- 步骤4.打开手机蓝牙，将手机靠近ESP32C3SuperMini，扫描设备并连接MyESP32设备



- 步骤 5.打开 LightBlue 应用程序并单击Bonded选项卡



- 步骤 6.单击MyESP32旁边的CONNECT



- 步骤 7.单击最底部显示“可读”、“可写”的部分

- 步骤 8.在数据格式下拉菜单下，选择UTF-8 字符串

- 步骤 9.在“ WRITTEN VALUES ”下键入“Hello” ， 然后单击“WRITE”

您将在 Arduino IDE 的串行监视器上看到文本字符串“Hello”输出

疑难解答

更多问题及有趣的应用，请访问论坛 或加入QQ技术交流群：522420541